



Affirmed Networks GTP Proxy Administration Guide

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- Product Overview
- GTP Proxy Configuration
- Performance Statistics

Affirmed Networks Inc.

35 Nagog Park
Acton, MA 01720
USA

Tel: +1-978-268-0800

www.affirmednetworks.com

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Preface

This guide describes an overview of the MCC GPRS Tunneling Protocol (GTP) Proxy and includes feature descriptions, interfaces, network deployments, and statistics.

***Note:** The Affirmed Networks Mobile Content Cloud™ (MCC) is the mobile services software solution that operates on virtual platforms such as blade systems.*

Audience

This guide is intended for network operators (sometimes referred to as Users or Operators in this guide), installation team staff, and other qualified service personnel responsible for provisioning and monitoring the system. The audience should be familiar with Long Term Evolution (LTE), 3G or 4G networks.

Release Information

See the Release Notes for important information, requirements, limitations, and restrictions that apply to this release.

How to Use This Guide

Read the following chapters and appendices in the order in which they appear:

Chapter/Appendix	Describes
Chapter 1, GTP Proxy Overview	An overview of the GTP Proxy including supported interfaces, services, network deployments, and call flows.
Chapter 2, MCC GTP Proxy Configuration	A description of the tasks used to configure the GTP Proxy.
Chapter 3, Statistics and Alarms	The GTP Proxy performance statistics viewable through the MCC CLI and the Acuitas Service Management System and alarms.

References and Specifications

The system supports the following references and specifications for the GTP Proxy:

- 3GPP TS 09.60 V7.10.0 (2002-12): *General Packet Radio Service (GPRS); GPRS Tunneling Protocol (GTP) across the Gn and Gp interface*
- 3GPP TS 23.401 V9.3.0 (2009-12): *General Packet Radio Service (GPRS) enhancements for Evolved Universal Terrestrial Radio Access Network (E-UTRAN) access*
- 3GPP TS 29.060 V9.3.0 (2010-06): *General Packet Radio Service (GPRS); GPRS Tunneling Protocol (GTP) across the Gn and Gp interface*
- 3GPP TS 29.274 V9.3.0 (2010-06): *3GPP Evolved Packet System (EPS); Evolved General Packet Radio Service (GPRS) Tunneling Protocol for Control plane (GTPv2-C); Stage 3*

Conventions

This guide uses the following conventions, when applicable:

Description	Convention	Examples
Command names, keywords, dialog boxes, buttons, icons, menus	Times New Roman bold font	Use configure_ha to configure... Enter y . To continue, press Enter .
Selecting a menu item	Menu => Option	Select Security => Group Management .
Variable parameters for which you supply values	<i><courier bold italic blue font></i>	configure_dr convert slave --masterhost <host IP>
User input	Courier bold blue font	cd /opt/Affirmed/NMS/bin
Screen messages, system output, sample source code	Courier regular blue font	Checking for License...
Required keywords and arguments in square brackets	[]	cluster [id] node [id] reboot
Optional keywords and arguments in curly brackets	{ }	{cdr-file-name}
Keywords separated by the pipe symbol. Requires you to specify only one item from the set.		Is this correct? [y n]
Keywords representing a single element in angle brackets	< >	<i><host name></i>
Asterisk after a CLI object indicates a required object.	*	name*
Variables, book and chapter titles, file path names, new terms, and emphasized text	<i>Times New Roman italic font</i>	See the <i>Acuitas User's Guide</i> . <i>Workflow</i> is an application running on the system.

Tip: Helpful suggestions.

Note: Additional information that may apply to the subject text.



CAUTION: Proceed carefully to avoid possible equipment damage or data loss.

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GTP Proxy Overview

This section contains an overview of the GPRS Tunneling Protocol (GTP) Proxy:

- [Product Overview \(page 1-2\)](#)
- [GTP Proxy Interfaces \(page 1-5\)](#)
- [PGW Selection \(page 1-6\)](#)
- [Workflow Profiles \(page 1-7\)](#)
- [Subscriber Analyzer Keys \(page 1-8\)](#)
- [Online Charging \(page 1-10\)](#)
- [Offline Charging \(page 1-11\)](#)
- [Charging Data Records \(page 1-11\)](#)
- [GTP Proxy Network Deployment Call Flows \(page 1-12\)](#)
- [APN Mapping \(page 1-18\)](#)
- [Path Management \(page 1-19\)](#)
- [Geographic Redundancy \(page 1-22\)](#)

Product Overview

The MCC GTP Proxy is a stand-alone virtualized, software-based service that acts as a GTP endpoint between the SGW and PGW in a 4G network and between the SGSN and GGSN in a 2G/3G network. The GTP Proxy provides stateful termination and relay of both GTP-C and GTP-U traffic from an SGW to a PGW via the S5/S8 interface or from an SGSN to a GGSN via the Gn/Gp interface.

The GTP Proxy solution supports the following services and features:

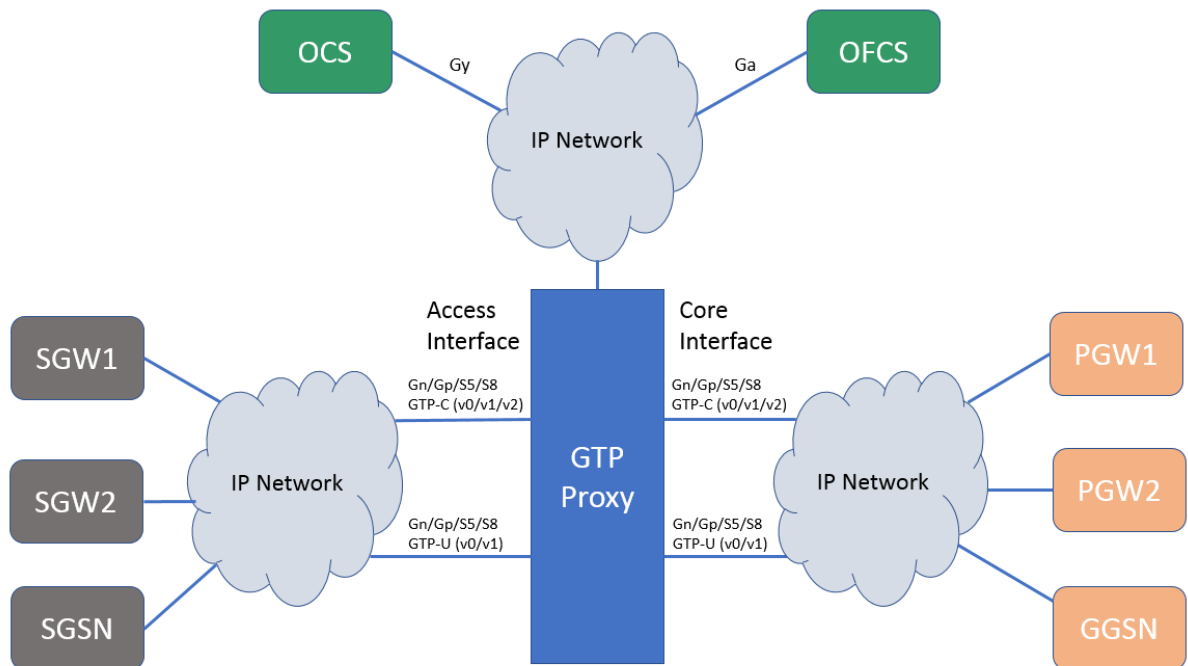
- Masks the home Public Land Mobile Network (HPLMN) topology from the roaming partner and only exposes the GTP Proxy's IP addresses (one IP address for the control plane and one IP address for the user plane)
- Handles incoming GTP-C messages from the visited Public Land Mobile Network (VPLMN) and selects and routes messages to the correct PGW using a standard or weighted round-robin algorithm from a configured list of PGWs/GGSNs
- Switches tunnels between access and core interfaces
- Passes through all GTP information elements (IEs) unchanged where possible
- Supports the GTPv0-C, GTPv1-C, GTPv2-C, GTPv0-U, and GTPv1-U GPRS tunneling protocols as defined in 3GPP TS 23.401, 3GPP TS 29.060 Version 9.3, 3GPP TS 29.274, and 3GPP TS 09.60 Version 7.10.0
- Supports offline CDRs (Local/Ga) for use in billing and accounting
- Supports the Gy online charging interface, which is used to authorize and manage a subscriber's quota and report data usage
- Supports Path Management procedures
- Supports APN mapping
- Supports Geographic Redundancy for GTP Proxy sessions
- Supports N-way local Redundancy for GTP Proxy sessions
- Supports GGSN/PGW selection and Workflow selection based on Subscriber Analyzers keys
- Supports GTP Proxy call performance statistics
- Supports alarms and protocol logging

Note: The total number of concurrent GTPv0 sessions supported on the GTP Proxy is 32000.

Note: For a list of all the supported GTP IEs on the GTP Proxy, see the *S5/S8 Interface* and *Gn/Gp Interface* chapters in the *Affirmed Networks Gateway Administration Guide*.

Figure 1-1 illustrates the position and role of the network components in a GTP Proxy deployment along with the network interfaces.

Figure 1-1 GTP Proxy Network Deployment



Configuration and Management

Network Operators can use the Command Line Interface (CLI) or the Acuitas Service Management System to configure and manage the GTP Proxy.

Command Line Interface – The CLI enables Operators to enter commands to perform specific tasks to configure, control, query, and monitor the GTP Proxy. See the *Affirmed Networks CLI Reference Guide* for details.

Acuitas Service Management System – The Acuitas Service Management System enables network Operators to manage the GTP Proxy using a standard, web-based management interface. Using Acuitas, Operators can monitor faults, alarms, performance statistics, provision and configure services, and perform software updates on any system. The Affirmed Networks Acuitas Server runs on a Linux (Red Hat) based operating system, while the client is accessible from a Firefox browser running on Windows 7 or Windows XP Professional. See the *Acuitas User's Guide* and *Acuitas System Administration Guide* for details.

For information on GTP Proxy performance statistics viewable through the MCC CLI and the Acuitas Service Management System, see [Chapter 3, Statistics and Alarms](#). For more information on performance statistics, see *Acuitas Performance Statistics*.

GTP Proxy Interfaces

The GTP Proxy supports the following network interfaces.

Table 1-1 GTP Proxy Network Interfaces

Interface	Description
Access Interface (S5/S8/Gn/Gp)	Connects the SGW/SGSN to the GTP Proxy using the General Packet Radio Service (GPRS) tunneling protocol (GTP-Cv0, GTP-Cv1,GTP-Cv2) for the control plane interface and using GTP-U (v0, v1) for the user plane interface. The GTP Proxy switches tunnels between the access and core interfaces and passes through all GTP IEs unchanged where possible. The GTP Proxy discards unsupported IEs before forwarding the message.
Core Interface (S5/S8/Gn/Gp)	Connects the GTP Proxy to the PGW/GGSN using GTP-C (v0, v1,v2) for the control plane interface and using GTP-U (v0,v1) for the user plane interface. The GTP Proxy switches tunnels between the access and core interfaces and passes through all GTP IEs unchanged where possible. The GTP Proxy discards unsupported IEs before forwarding the message. The GTP Proxy maintains its own restart counter for path management and replaces the restart counter sent by the SGW/SGSN peer with the locally stored restart counter before forwarding the message.
Gy Interface	Connects the GTP Proxy to the Online Charging System (OCS) and is used to authorize and manage a subscriber's quota and report data usage. The Gy interface uses the Diameter Credit Control Application (DCCA) as defined in RFC 4006.
Ga Interface	Connects the GTP Proxy to the Offline Charging System (OFCS) using GTP' and transports Abstract Syntax Notation 1 (ASN.1) encoded CDR files.
Local Interface	Used to store offline charging CDRs locally in both comma-separated values (CSV) and Abstract Syntax Notation 1 (ASN.1) format.

PGW Selection

When the GTP Proxy receives a session setup request for a user equipment (UE), it classifies the UE according to configured Subscriber Analyzer keys, such as the International Mobile Subscriber Identity (IMSI) range, Access Point Name (APN) or UE location. The UE's classification resolves to a list of locally configured PGWs/GGSNs. The GTP Proxy selects the appropriate PGW/GGSN from the list using a standard or weighted round-robin algorithm and forwards the session setup request. A configured PGW/GGSN will only be used for selection if its link status is up and it supports the GTP version and IP address type that the session is using. To select a GGSN/PGW based on Subscriber Analyzer keys, the `analyzer-type` must be set to `peer-selector`.

***Note:** For a list of supported Subscriber Analyzer keys used for PGW/GGSN selection, see [Subscriber Analyzer Keys \(page 1-8\)](#).*

Example

If either KEY1 or KEY2 matches, the configured `pgw-peer-list` will be used.

With KEY1, the `apn-name`, `gtp-proxy-name` and `serving-nw-mcc` have to match to use the configured `pgw-peer-list`.

```
an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer ANALYZE1
key KEY1 apn-name internet.mnc401.mcc201.gprs gtp-proxy-name PGWPROXY
serving-nw-mcc 310
```

With KEY2, only the `imsi-range` needs to match to use the configured `pgw-peer-list`.

```
an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer ANALYZE1
key KEY2 imsi-range high 900000000000000 low 800000000000000
```

The following CLI command is used to associate KEY1 and KEY2 with the configured PGW peer list for the Subscriber Analyzer `ANALYZE1`.

```
an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer ANALYZE1
analyzer-type peer-selector geo-redundant group GEOGRP1 priority 1 key
KEY1 key KEY2 peer-list pgw-peer-list
```


Workflow Profiles

Workflow defines the treatment of a service flow across various services. The Workflow Subscriber Analyzer is used to select a Workflow Control and Data profile for a subscriber based on configured keys, such as APN name or IMSI range. The selected Control profile determines the Gy and Ga interface parameters as well as APN mapping parameters to be used for the subscriber. To select a Workflow Control and Data profile based on Subscriber Analyzers keys, the `analyzer-type` must be set to `workflow-profile`.

For Workflow configuration information, see [Configuring Workflow Components and Services \(page 2-19\)](#).

Example

```
an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer ANALYZE1
analyzer-type workflow-profile key KEY1 apn-name
internet.mnc401.mcc201.gprs action control-profile CTRLPROF
data-profile DATAPROF
```

The selected Workflow Control profile determines the online (Gy interface) and offline (Ga interface) charging parameters as well as APN mapping parameters to apply to the UE. The APN name reported to external servers, such as the OCS, will be the APN name derived from the configured `apn-reporting` parameter.

```
an3000-mcm-slot7cpu1(config)# workflow control-profile CTRLPROF
default-online-interface OCSCHRG
an3000-mcm-slot7cpu1(config)# workflow control-profile CTRLPROF
default-offline-interface GTPPCHRG_PGW
an3000-mcm-slot7cpu1(config)# workflow control-profile CTRLPROF
apn-mapping-control explicit mapped-apn-name internet apn-reporting
original
```

Note: For a list of supported Subscriber Analyzer keys used to select a Workflow Control and Data profile, see [Subscriber Analyzer Keys \(page 1-8\)](#).

Subscriber Analyzer Keys

You can configure multiple keys, each identified by name. If any key matches, the Subscriber Analyzer is selected. A key can specify multiple triggers, all of which must match for a key to match.

```
an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer
<subscriber-analyzer-name> key <key-name> <your-list-of-keys>
<key-values>
```

Example

```
an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer ANALYZE1
key KEY2 imsi-range high 900000000000000 low 800000000000000
```

The Subscriber Analyzer supports the following analyzed key parameters used for PGW/GGSN and Workflow selection.

Parameter	Description
apn-name	Specify the Access Point Name. <i>Note: If the APN name is used as part of a key and APN mapping has been implemented for the subscriber resulting in a new APN name, the final mapped APN name will be used when specifying the key.</i>
cell-global-id	Specify the Cell global identifier.
eutran-cell-global-id	Specify the E-Utran cell global identifier.
gtp-proxy-name	Specify the GTP Proxy service name on which the session setup request is received.
home-plmnid	Specify the subscriber's home PLMN ID, 5 or 6 digits.
imei-range	Specify the International Mobile Equipment Identity range.
imsi-range	Specify the International Mobile Subscriber Identity range.
rat-type	Select the Radio Access Type: ■ UTRAN ■ GERAN ■ WLAN ■ GAN ■ HSPA Evolution ■ EUTRAN
roaming-user	Specify whether user is a roaming user. ■ true ■ false

Parameter	Description
routing-area-id	Specify the Routing Area Identifier (RAI): ■ location-area-code ■ plmnid – RAI PLMN ID, 5 or 6 digits ■ routing-area-code
service-area-id	Specify the Service Area Identifier (SAI). ■ location-area-code ■ plmnid – SAI PLMN ID, 5 or 6 digits ■ service-area-code
serving-nw-mcc	Specify the serving network's Mobile Country Code (MCC) from which the session setup request is received. Note the following: ■ For a GTPv2 session, the serving network's MCC is the first three digits of the PLMN ID received in the Serving Network IE field present in the Create Session Request message. ■ For a GTPv1 session, the serving network's MCC is the first three digits of the PLMN ID received in the Routing Area Identity field of the Create PDP Context Request message. The Routing Area Identity field is an optional field and may not be available within the Create PDP Context Request message in which case the GTP Proxy cannot use this field as a key.
serving-nw-plmnid	Specify the serving network's PLMN ID, 5 or 6 digits.
tracking-area-id	Specify the Tracking Area Identifier (TAI). ■ plmnid – TAI PLMN ID, 5 or 6 digits ■ tracking-area-code
ue-ip-address	Enter the User Equipment static IP Address.

Online Charging

The GTP Proxy supports the Gy online charging interface, which is used to authorize and manage a subscriber's quota and report data usage. The Gy interface uses the Diameter Credit Control Application (DCCA) as defined in RFC 4006.

The GTP Proxy supports the following online charging functions:

- Multi-service rating-group level credit control
- Quota requesting and usage reporting on time, volume, and unit based thresholds
- Quota Holding Time (QHT) and Quota Consumption Time (QCT)
- Validity time
- Tariff time change reporting
- Envelope reporting
- Event trigger on rating condition change
- Network-initiated credit reauthorization from the OCS
- Network-initiated session termination from the OCS
- Advice of charge

The selected Workflow Control profile determines the Gy interface parameters to be used for the UE. When configuring the online charging interface associated with a Workflow Control profile, Affirmed Networks recommends configuring an initial quota to allow data traffic to pass through the GTP Proxy if it is received before the GTP Proxy has contacted the OCS. Alternatively, Operators have the option to disable online charging for this initial traffic.

```
an3000-mcm-slot7cpu1(config)# service-construct interface
online-charging OCSCHRG interface-compliance 3gppVersion2
rating-group-initial-quota 1000000 charge-rating-group-initial-quota
false
```

See the *Affirmed Networks Gy Interface Specification* for information on configuring the Gy interface and charging components, such as rating groups, billing plans and charging instances.

Offline Charging

The GTP Proxy supports the Ga offline charging interface, which is used to transport ASN.1 encoded CDR files from the GTP Proxy to an Offline Charging System (OFCS). The Ga interface uses a GTP-based protocol (called GTP', GTP prime or GTPP), with modifications to support CDRs. See the *Affirmed Networks Charging Data Records Specification* for information on configuring the Ga interface.

Charging Data Records

The GTP Proxy generates offline Charging Data Records (CDRs) for use in billing and accounting. The GTP Proxy stores PGW CDRs locally in both CSV and ASN.1 format. You can configure the system to forward CDRs to the Offline Charging System (OFCS) via the Ga interface using GTP'. The selected Workflow Control profile determines the Ga interface parameters to be used for the UE.

The GTP Proxy uses a locally generated charging ID in PGW CDRs and sends this charging ID in the Create Session Response/Create PDP Context Response to the SGW/SGSN. The GTP Proxy ignores/discards the charging ID received from the PGW/GGSN.

The GTP Proxy does not generate GGSN CDRs and instead generates PGW CDRs for both PGW and GGSN GTP Proxy sessions.

When used for a GTP Proxy session, the following PGW CDR fields are filled as follows:

- PGW Address – Contains the GTP Proxy IPv4 address.
- PGW Address (IPv6) – Contains the GTP Proxy IPv6 address (if available).
- Charging ID – Contains the Charging ID generated by the GTP Proxy.
- Serving Node Address – Contains the SGW/SGSN IP address.

For more information on CDRs, see the *Affirmed Networks Charging Data Records Specification*.

GTP Proxy Network Deployment Call Flows

This section provides examples of the GTP Proxy network deployment call flows.

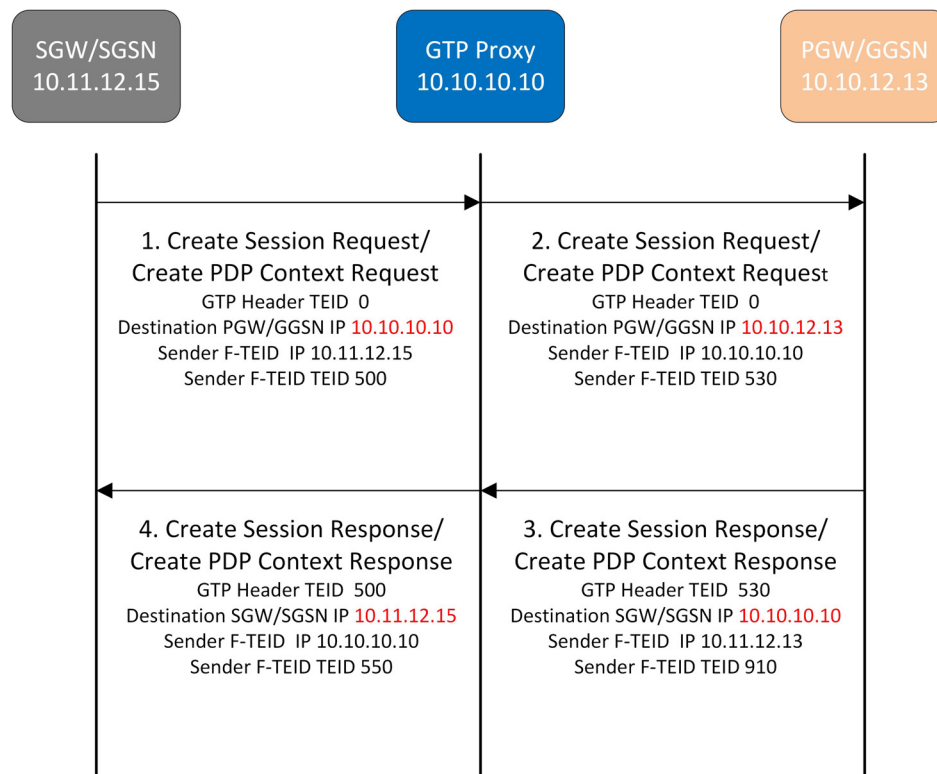
- [Session Set Up \(page 1-13\)](#)
- [Session Deletion \(page 1-14\)](#)
- [Modify Bearer Request for ULI Change \(page 1-15\)](#)
- [Intra-LTE Handover with SGW Change \(page 1-16\)](#)
- [4G/LTE to 3G Handover \(page 1-17\)](#)

Session Set Up

Figure 1-2 illustrates an example of the call flow for when the GTP Proxy receives a Create Session Request/Create PDP Context Request.

If the GTP Proxy receives a session setup request for an existing session, it will deactivate the session locally and send a Delete Session/Delete PDP Context Request to the PGW/GGSN hosting the session. The GTP Proxy will re-establish the session after selecting a PGW/GGSN from the configured list.

Figure 1-2 GTP Proxy Create Session Request/Create PDP Context Request



- 1 The SGW/SGSN sends a Create Session Request/Create PDP Context Request to the GTP Proxy using the GTP Proxy's access interface IP address as the Destination PGW/GGSN IP address.

Note: If an SGW/SGSN sends an Indication Flags IE with the "Piggybacking Supported" flag set, the GTP Proxy unsets the flag and forwards the message to PGW/GGSN.

- 2 The GTP Proxy selects a PGW/GGSN from the configured list, replaces the Sender F-TEID for control plane with the core interface's F-TEID and forwards the message to the PGW/GGSN.

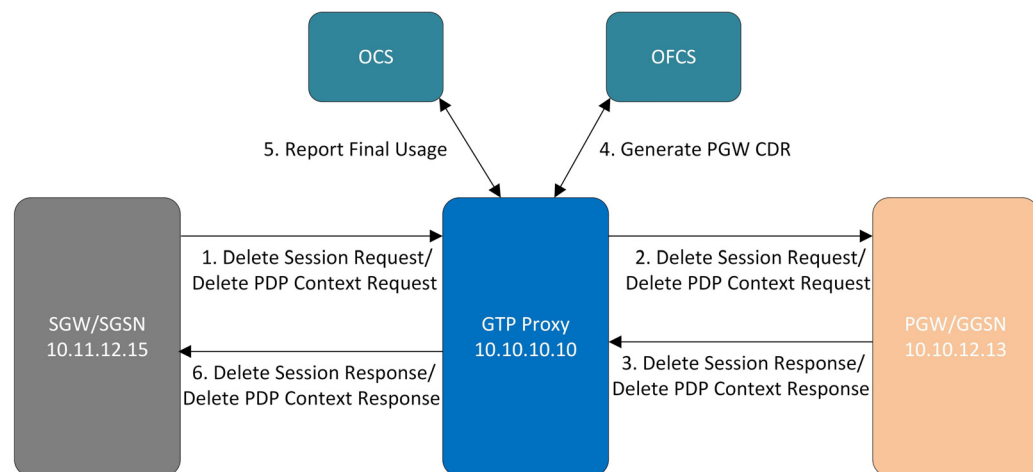
- 3 The PGW/GGSN responds with a Create Session Response/Create PDP Context Response using the GTP Proxy's core interface IP address as the Destination SGW/SGSN IP address.
- 4 The GTP Proxy replaces the sender F-TEID for control plane with the access interface's F-TEID and forwards the Create Session Response/Create PDP Context Response to the SGW/SGSN.

Note: The GTP Proxy does not enforce any QoS parameters.

Session Deletion

Figure 1-3 illustrates an example of the call flow for when the GTP Proxy deletes a session using the Delete Session Request/Delete PDP Context Request.

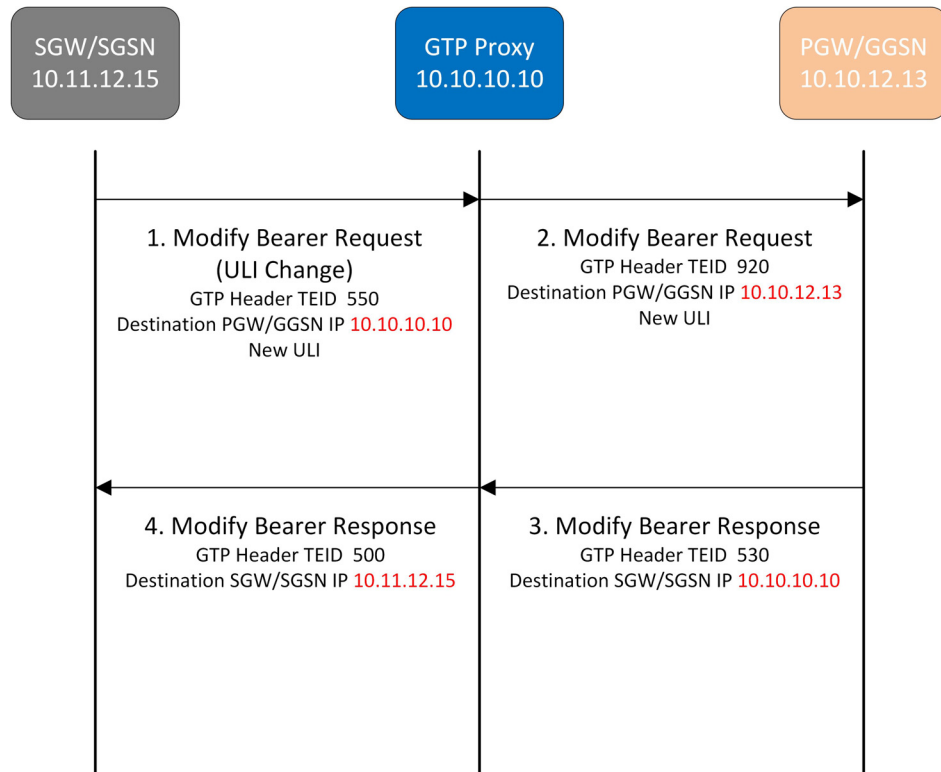
Figure 1-3 GTP Proxy Delete Session Request/Delete PDP Context Request



Modify Bearer Request for ULI Change

Figure 1-4 illustrates an example of the call flow for when the GTP Proxy receives a Modify Bearer Request message for a User Location Information (ULI) change.

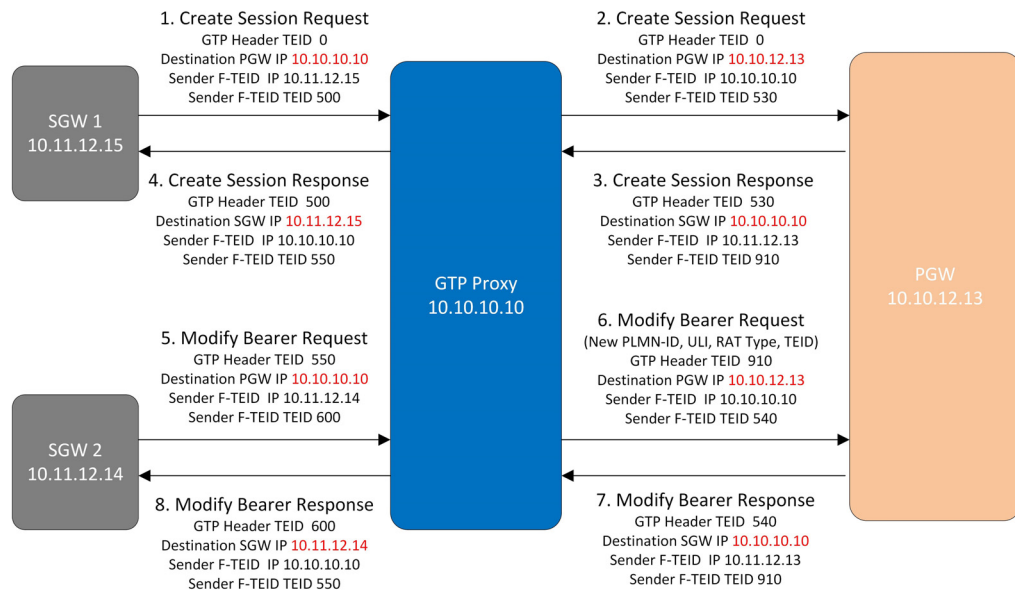
Figure 1-4 GTP Proxy Modify Bearer Request for ULI Change



Intra-LTE Handover with SGW Change

Figure 1-5 illustrates an example of the call flow for when the GTP Proxy handles an intra-LTE handover with SGW Change.

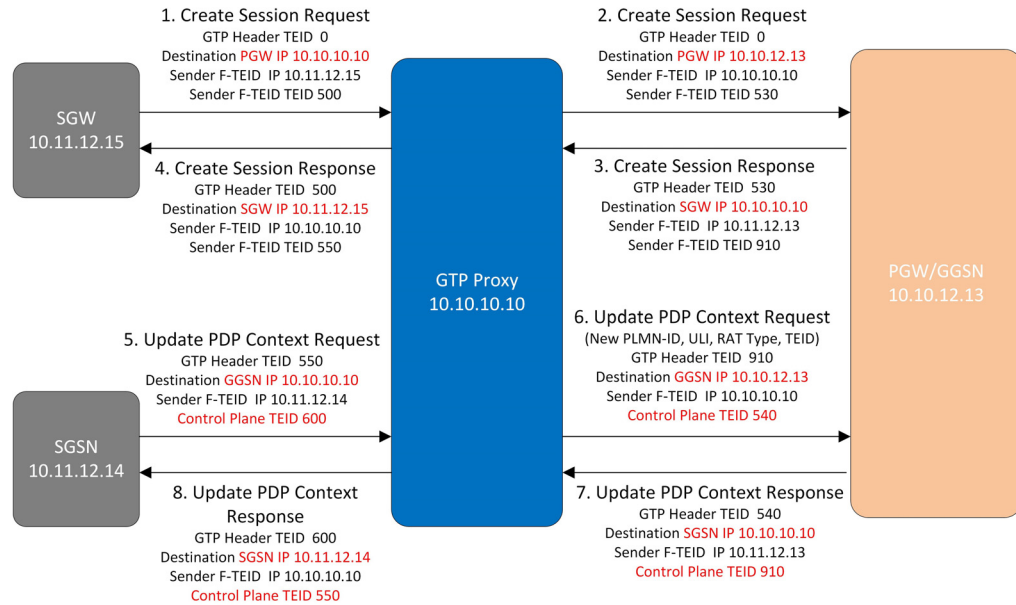
Figure 1-5 GTP Proxy Intra-LTE Handover with SGW Change



4G/LTE to 3G Handover

Figure 1-6 illustrates an example of the call flow for when the GTP Proxy handles a 4G/LTE to 3G handover.

Figure 1-6 GTP Proxy 4G/LTE to 3G Handover



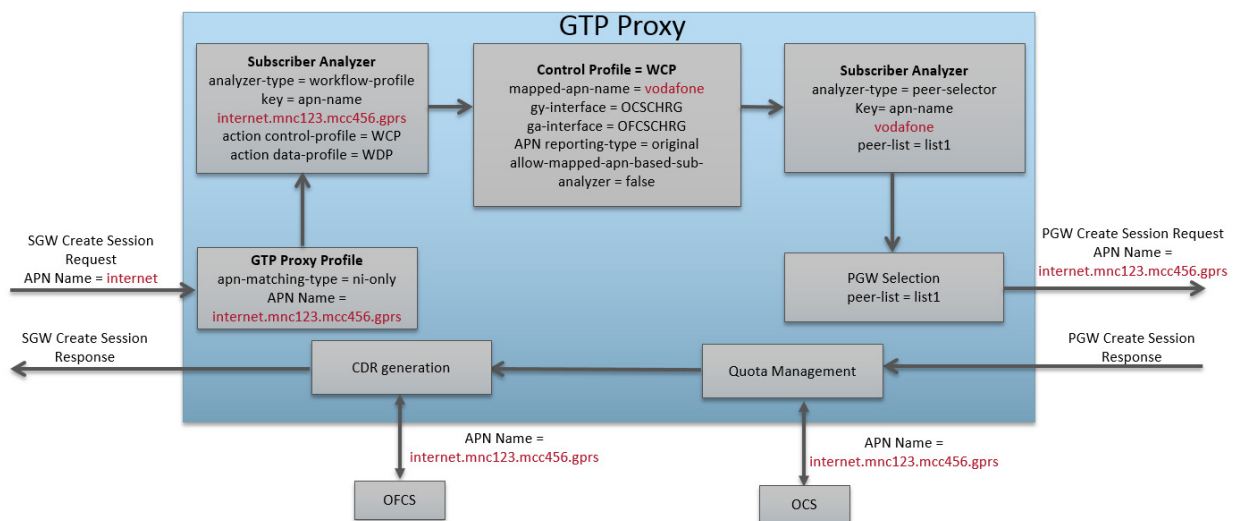
APN Mapping

The GTP Proxy supports the ability to map the APN name received in a Create Session/Create PDP Context Request to a different APN name.

When the GTP Proxy receives a session setup request, the Subscriber Analyzer is used to select a Workflow Control profile for the UE. The Workflow Control profile can be associated with an APN name to which the received APN name is mapped. Using the `reporting-type` parameter under the Workflow Control profile, Operators can specify the APN name that will be forwarded to the PGW/GGSN and reported on the Gy and Ga interfaces. See [Configuring Workflow Components and Services \(page 2-19\)](#) for information on configuring APN mapping.

Figure 1-7 illustrates the call flow for when the GTP Proxy receives a session setup request and APN mapping is used to map the APN name received in the Create Session/Create PDP Context Request to a different APN name. It also shows how the mapped APN is reported to the PGW/GGSN and external devices, such as the OCS and OFCS.

Figure 1-7 APN Mapping with GTP Proxy



Path Management

When Path Management procedures are initiated, the GTP Proxy transmits GTP-C/GTP-U Echo Requests on access and core interfaces for both static and dynamic peers if the `initiate-echo-requests` parameter is set to `true` for these interfaces. The GTP Proxy learns of dynamic peers upon receipt of the Create Session/Create PDP Context Requests and Responses. However, the GTP Proxy does not initiate GTP-C Echo Request messages towards GTP peers added based on the source IP address included in the IP protocol header in incoming Create Session Request/Create PDP Context messages. Instead, the GTP Proxy only initiates GTP-C Echo Requests towards:

- Configured peers (Static GTP peers)
- GTP peers added based on the IP address encoded in the Sender F-TEID for Control Plane IE of the Create Session Request/Create PDP Context messages.

The GTP Proxy service maintains its own restart counter for path management and replaces the restart counter sent by the SGW/SGSN peer with the locally stored restart counter before forwarding the message to the PGW/GGSN.

If the GTP Proxy detects that the restart counter of an SGW/SGSN has changed, it will cleanup sessions on the PGW/GGSN corresponding to the failed SGW/SGSN. If the GTP Proxy detects that the restart counter of a PGW/GGSN has changed, it will cleanup sessions on the SGW/SGSN corresponding to the failed PGW/GGSN. The sessions will be cleaned up in a paced manner.

Note: The system initiates a subscriber deactivation based on the configured `subscriber-bulk-deactivation-rate-control` parameters. For more information, see [Configuring the GTP Proxy Profile \(page 2-13\)](#).

This section provides examples of path management call flows.

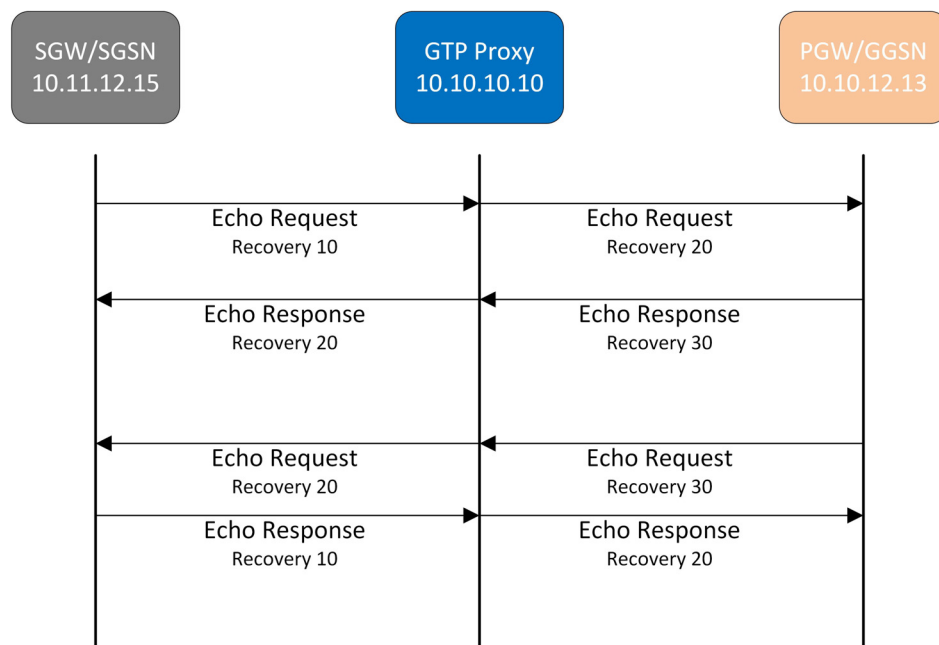
- [Path Management Echo Request \(page 1-20\)](#)
- [PGW/GGSN Path Failure \(page 1-21\)](#)
- [SGW/SGSN Path Failure \(page 1-21\)](#)

Path Management Echo Request

Figure 1-8 illustrates an example of the call flow for when the GTP Proxy initiates path management procedures for peers associated with a GTP Proxy interface. The GTP Proxy sends Echo Request messages to the peer(s) and waits for a response.

The `initiate-echo-requests` parameter must be set to `true` for the GTP Proxy to transmit Echo Requests on access and core interfaces.

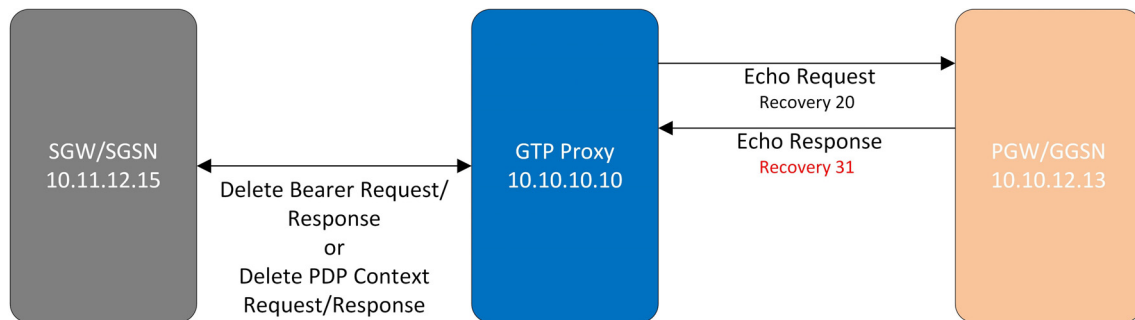
Figure 1-8 Path Management ECHO Request



PGW/GGSN Path Failure

Figure 1-9 illustrates an example of the call flow for when the GTP Proxy detects a path failure after initiating path management procedures for a PGW/GGSN peer.

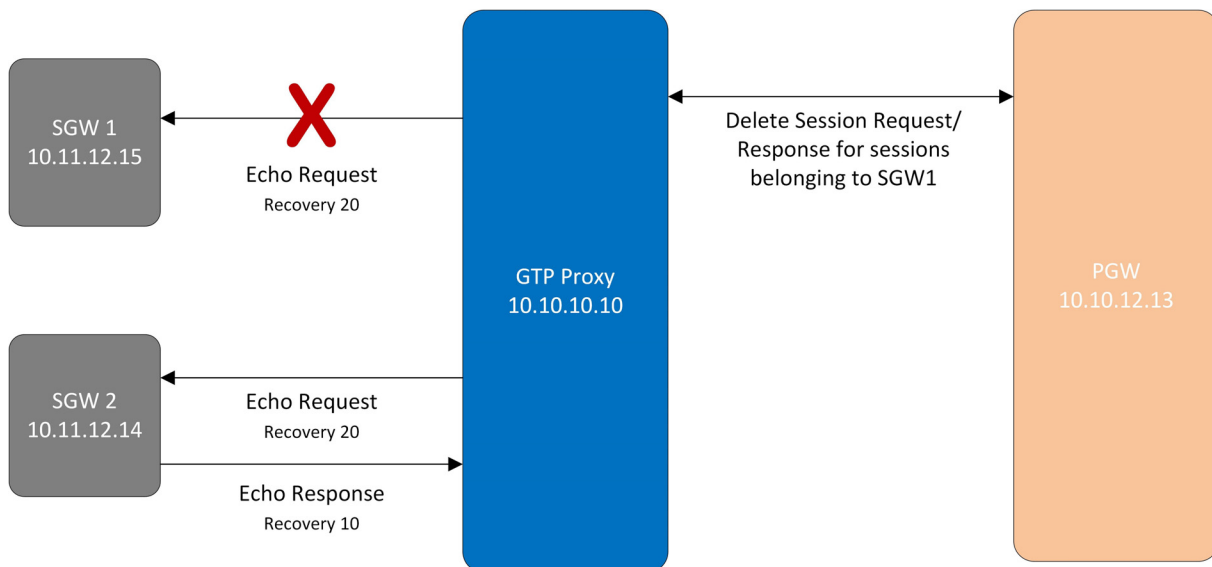
Figure 1-9 PGW/GGSN Path Failure



SGW/SGSN Path Failure

Figure 1-10 illustrates an example of the call flow for when the GTP Proxy detects a path failure after initiating path management procedures for an SGW/SGSN peer.

Figure 1-10 SGW/SGSN Path Failure



Geographic Redundancy

The GTP Proxy supports Local and Geographic Redundancy for GTP Proxy sessions. The Geographic Redundancy solution is based on a collaborative system of two geographically-separated network elements functioning as standby partners for each other. A Geographic Redundancy group has a designated primary and a designated secondary system. Each group represents a set of services with an active representation on the primary node and a standby representation on the secondary node. If the active node goes out of commission, the services on the standby node can be enabled so these services can resume.

It is recommended that you use the Geographic Redundancy Manager application in the Acuitas Service Management System to configure the Geographic Redundancy group, service endpoint, and server parameters. For more information on configuring Geographic Redundancy, see the *Acuitas User's Guide*.

Only the Gy interface is supported for network health check on the GTP Proxy. Network health check must be enabled on the Gy interface using both shared and private loopback IP addresses. For information on configuring Geographic Redundancy health checks on the Gy interface, see the *Affirmed Networks Gateway Administration Guide*.

MCC GTP Proxy Configuration

This section provides information on configuring the MCC GTP Proxy.

- [Configuring Network Contexts and Loopback IP Addresses \(page 2-2\)](#)
- [Configuring the GTP Proxy Service \(page 2-4\)](#)
- [Configuring the Access and Core Interfaces \(page 2-6\)](#)
- [Configuring a GTP Interface Profile \(page 2-9\)](#)
- [Configuring GTP Proxy Peers \(page 2-11\)](#)
- [Configuring GTP-C Endpoint Parameters \(page 2-12\)](#)
- [Configuring the GTP Proxy Profile \(page 2-13\)](#)
- [Configuring APNs \(page 2-15\)](#)
- [Configuring Home and Roaming PLMN IDs \(page 2-16\)](#)
- [Configuring a List of PGW/GGSNs \(page 2-17\)](#)
- [Configuring Workflow Components and Services \(page 2-19\)](#)

See the *Affirmed Networks CLI Reference Guide* for a complete listing of CLI commands used in this section.

See the *Affirmed Networks Gateway Administration Guide* for more information about the CLI commands used in this section.

For information on configuring the following offline charging system (OFCS) parameters, see the *Affirmed Networks Charging Data Records Specification*.

- Storing Charging Data Records (CDRs) in a local file.
- Configuring the local and Ga offline charging interface.

Configuring Network Contexts and Loopback IP Addresses

Each network context maintains its own instance of configuration settings, IP interface parameters and routing tables. MCC supports IPv4 and IPv6 loopback addresses within each network context. The GTP Proxy is flexible when configuring network contexts. Some options include:

- One network context for all four interfaces (Access GTP-C, Access GTP-U, Core GTP-C, Core GTP-U)

Note: *If the access and core interfaces share the same network context, the access GTP-C loopback IP address must match the core GTPv2-C loopback IP address.*

- Two network contexts
 - Access GTP-C/GTP-U
 - Core GTP-C/GTP-U
- Four network contexts
 - Access GTP-C
 - Access GTP-U
 - Core GTP-C
 - Core GTP-U

The configuration procedures provided in this section show how to configure two network contexts for the GTP Proxy, one for the access interfaces and another for the core interfaces.

- Access Network Context – Connects the GTP Proxy to the SGW/SGSN. See [Configuring the Access Network Context \(page 2-3\)](#).
- Core Network Context – Connects the GTP Proxy to the PGW/GGSN. See [Configuring the Core Network Context \(page 2-3\)](#).

For more information on network contexts, see the *Affirmed Networks Gateway Administration Guide*.

Configuring the Access Network Context

Create the access network context and configure the access GTP-U and GTP-C interface loopback IP addresses.

```
an3000-mcm-slot7cpu1(config)# network-context <nw-context-name>
loopback-ip <loopback-name> ip-address <ip-address> gtp-u [yes|no]
```

where

Parameter	Description
network-context	Specifies the name of the access network context.
loopback-ip	Specifies the name of the local loopback IP address associated with the access network context.
ip-address	Specifies the IP address of this loopback object. Can be IPv4/IPv6.
gtp-u	Specifies whether to use this loopback address for GTP-U. ■ yes ■ no <i>Note: Specify yes when configuring the Access GTP-U interface loopback IP address.</i>

Configuring the Core Network Context

Create the core network context and configure the core GTP-U and GTP-C interface loopback IP addresses.

```
an3000-mcm-slot7cpu1(config)# network-context <nw-context-name>
loopback-ip <loopback-name> ip-address <ip-address> gtp-u [yes|no]
```

where

Parameter	Description
network-context	Specifies the name of the core network context.
loopback-ip	Specifies the name of the local loopback IP address associated with the core network context.
ip-address	Specifies the IP address of this loopback object. Can be IPv4/IPv6.
gtp-u	Specifies whether to use this loopback address for GTP-U. ■ yes ■ no <i>Note: Specify yes when configuring the core GTP-U interface loopback IP address.</i>

Configuring the GTP Proxy Service

To enable GTP Proxy functionality on the MCC, Operators configure a GTP Proxy service.

Note: GTP Proxy services cannot share the same GTP-C endpoint addresses or loopback IP addresses with each other or with any other service running on the MCC.

Use the following CLI command to configure the GTP Proxy service.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway gtp-proxy
<gtp-proxy-name>
```

The following table describes the configurable parameters.

Parameter	Description
gtp-proxy	Specifies the name of the GTP Proxy service.
admin-state	Specifies whether to enable or disable the administrative state for this service. ■ enabled ■ disabled Note: Disable the admin-state of the GTP Proxy service before making modifications.
access-gtpc-interface	Specifies the access GTP-C interface name. See Configuring the Access and Core Interfaces (page 2-6) .
access-gtpu-interface	Specifies the access GTP-U interface parameters. See Configuring the Access and Core Interfaces (page 2-6) .
core-gtpc-interface	Specifies the core GTP-C interface parameters. See Configuring the Access and Core Interfaces (page 2-6) .
core-gtpu-interface	Specifies the core GTP-U interface parameters. See Configuring the Access and Core Interfaces (page 2-6) .
geo-redundant-group	Specifies the Geographic Redundancy Group name. For information, see Geographic Redundancy (page 1-22) .
gtp-proxy-endpoint-parameters	Configure the GTP Proxy endpoint parameters. See Configuring Workflow Components and Services (page 2-19) .
gtp-proxy-profile	Specifies the GTP Proxy Profile. See Configuring the GTP Proxy Profile (page 2-13) .
operation-mode	Specifies the GTP Proxy operation mode. Currently, the only supported mode is <code>gtpc-and-gtpu</code> .

Parameter	Description
apn-list	Specifies the list of APN names. The APN(s) must already exist. See Configuring APNs (page 2-15)
home-plmnid-list	Specifies the list of PLMN IDs considered as home. The home PLMN-IDs must already exist. See Configuring Home and Roaming PLMN IDs (page 2-16)
roaming-plmnid-list	Specifies the list of PLMN IDs considered as roaming. The roaming PLMN-IDs must already exist. See Configuring Home and Roaming PLMN IDs (page 2-16)

Configuring the Access and Core Interfaces

Use the following CLI command to configure the access and core GTP-C/GTP-U interface parameters.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway gtp-proxy
<gtp-proxy-name> [access-gtpc-interface|access-gtpu-interface|
core-gtpc-interface|core-gtpu-interface]
```

Note: The network context and IPv4 and IPv6 addresses configured for the access GTP-C interface must match those configured for the GTP-C endpoint parameters.

The following table describes the configurable parameters.

Parameter	Description
gtp-proxy	Specifies the name of the GTP Proxy service.
admin-state	Specifies whether to enable or disable the administrative state for this service. ■ enabled ■ disabled
[access-gtpc-interface access-gtpu-interface core-gtpc-interface core-gtpu-interface] Configure the following parameters for each GTP Proxy interface.	
gtp-interface-profile	Specifies the name of the GTP Interface Profile to use for this interface. The GTP Interface Profile contains GTP-C/GTP-U data path management settings (timers and counts) used to maintain data path status. The GTP Interface Profile must already exist. See Configuring a GTP Interface Profile (page 2-9) .
network-context	Specifies the name of the network context to which this interface belongs. The Network Context must already exist. See Configuring Network Contexts and Loopback IP Addresses (page 2-2) .

Parameter	Description
<code>loopback-list</code>	<i>Applies to the <code>access-gtpc-interface</code>, <code>access-gtpu-interface</code>, and <code>core-gtpu-interface</code> only.</i> Specifies the names of the loopback IP addresses (associated with this network context) used for the interface. Can be IPv4 or IPv6. Note the following restrictions: ■ Once configured, GTP-C loopback IP addresses cannot be deleted from the GTP Proxy configuration. The GTP Proxy service configuration must be deleted and re-added to delete GTP-C loopback IP addresses. ■ Only one IPv4 and one IPv6 address can be configured in the <code>loopback-list</code> for the <code>access-gtpc-interface</code>. ■ The IPv4 and IPv6 addresses configured for the <code>access-gtpc-interface</code> must match the IPv4 and IPv6 addresses configured in <code>gtp-endpoint-parameters</code>. The loopback IP addresses and names must already be configured for the network context for which this interface belongs. See Configuring Network Contexts and Loopback IP Addresses (page 2-2).
<code>[gtpv1-loopback-list gtpv2-loopback-list]</code>	<i>Applies to the <code>core-gtpc-interface</code> only.</i> Specifies the names of the core GTPv1-C/GTPv2-C loopback IP addresses (associated with this network context) used for the interface. Can be IPv4 or IPv6. Note the following restrictions: ■ Once configured, GTP-C loopback IP addresses cannot be deleted from the GTP Proxy configuration. The GTP Proxy service configuration must be deleted and re-added to delete GTP-C loopback IP addresses. ■ Only one IPv4 and one IPv6 address can be configured in the <code>gtpv1-loopback-list</code> and <code>gtpv2-loopback-list</code>. ■ The IPv4 and IPv6 addresses configured in the <code>gtpv2-loopback-list</code> must be different from the IPv4 and IPv6 addresses configured for the <code>gtpv1-loopback-list</code>. ■ If you configure an IPv4 and/or IPv6 address for the <code>gtpv1-loopback-list</code>, then you must configure the same types (IPv4 and/or IPv6) of IP addresses for the <code>gtpv1-loopback-list</code>. ■ If the access and core interfaces share the same network context, the access GTP-C loopback IP address must match the core GTPv2-C loopback IP address. The loopback IP addresses and names must already be configured for the network context for which this interface belongs. See Configuring Network Contexts and Loopback IP Addresses (page 2-2).

Parameter	Description
<code>peer</code>	<i>Applies to the <code>access-gtpc-interface</code> and <code>core-gtpc-interface</code> only.</i> Specifies the names of the GTP-C peers associated with this interface. A peer associated with the <code>access-gtpc-interface</code> or <code>access-gtpc-interface</code> must have the same network context ID as the interface. The GTP-C Peer must already exist. See Configuring GTP Proxy Peers (page 2-11).

Configuring a GTP Interface Profile

The GTP Interface Profile contains GTP-C data path management settings (timers and counts) used to maintain data path status. The GTP Proxy access and core interfaces are associated with a GTP Interface Profile. See [Path Management \(page 1-19\)](#) for more information.

Use the following CLI command to configure a GTP Interface Profile used by GTP interfaces for GTP-C data path management.

```
an3000-mcm-slot7cpu1(config)# zone default gateway profile
gtp-interface-profile <gtp-interface-profile-name>
```

The following table describes the configurable parameters.

Parameter	Description
gtp-echo-interval	Specifies the amount of time (ms) to wait before initiating an Echo Request. When Path Management procedures are initiated, Echo requests are transmitted periodically according to the configured value of the gtp-echo-interval .
gtp-echo-response-timeout	Specifies the amount of time (ms) to wait before retransmitting an Echo Request.
gtp-echo-retransmit-count	Specifies the maximum number of Echo Request retransmission attempts (not including the initial Echo Request) to make before declaring a path down. Path failures are detected if there is no Echo Response received after exhausting the gtp-echo-retransmit-count retries. Path failures are also detected if the peer changes its Restart Counter value.
initiate-echo-requests	Specifies whether to send periodic Echo Requests to peers upon learning peer information. ■ true – (default) When peers are configured and associated with a GTP Proxy interface, the GTP Proxy begins path management procedures for each peer. ■ false – The GTP Proxy does not send any Echo Requests to the peer.
n3-count	Specifies the number of times the gateway retransmits a message if no acknowledgment is received. Affirmed Networks has the following recommendations for configuring the n3-count parameter. ■ Configure the N3 count on the core interface to be less than the N3 count configured on the SGSN/SGW to prevent the SGW/SGSN from timing out a procedure before the GTP Proxy detects the failure. ■ Configure the N3 count on the access interface to be less than the N3 count configured on the PGW/GGSN to prevent the PGW/GGSN from timing out a procedure before the GTP Proxy detects the failure.

Parameter	Description
<code>t3-timeout</code>	Specifies the length of time to wait for an acknowledgment for a message (ms). Affirmed Networks has the following recommendations for configuring the <code>t3-timeout</code> parameter. ■ Configure the T3 timeout value on the core interface to be less than the T3 timeout value configured on the SGSN/SGW to prevent the SGW/SGSN from timing out a procedure before the GTP Proxy detects the failure. ■ Configure the T3 timeout value on the access interface to be less than the T3 timeout value configured on the PGW/GGSN to prevent the PGW/GGSN from timing out a procedure before the GTP Proxy detects the failure.

Configuring GTP Proxy Peers

Operators can configure GTP Proxy peers (remote SGW/PGW) and associate those peers with the access or core GTP-C interface of a GTP Proxy service. A peer is identified by its IP address, network context ID, the version of the GTP-C protocol it supports and whether the peer is present in the same PLMN as the GTP Proxy. A peer associated with an access or core GTP-C interface must have the same network context ID as the interface.

Operators can configure the GTP Proxy to initiate path management procedures for these statically configured peers using the GTP Interface Profile. See [Configuring a GTP Interface Profile \(page 2-9\)](#) for more information.

- 1 Configure the peers for the access and core GTP-C interface.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway gtp-peer <peer-name>
gtpversion [gtpv0|gtpv1|gtpv2] is-present-in-local-plmn [false|true]
network-context <nw-context-name> peer-ip-address <ip-address>
```

- 2 Associate the peer with the interface by adding the peer to the list of GTP-C peers with which the interface communicates.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway gtp-proxy
<gtp-proxy-name> [access-gtpc-interface|core-gtpc-interface] peer
<peer-name>
```

Configuring GTP-C Endpoint Parameters

Use the following CLI command to configure the GTP-C endpoint parameters used for the control plane handling of signaling messages on the GTP Proxy. The SGW/SGSN uses the GTP-C endpoint parameter IP address to send session requests to the GTP Proxy.

Note: The network context and IPv4 and IPv6 addresses configured for the GTP-C endpoint parameters must match those configured for the `access-gtpc-interface`.

Note: GTP Proxy services cannot share the same GTP-C endpoint addresses or loopback IP addresses with each other or with any other service running on the MCC.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway gtp-proxy  
<gtp-proxy-name> gtp-proxy-endpoint-parameters
```

The following table describes the configurable parameters.

Parameter	Description
<code>network-context</code>	Specifies the name of the network context on which this GTP-C endpoint is anchored.
<code>ipv4-loopback</code>	Specifies the name of the IPv4 loopback IP address (associated with this network context) used for this GTP-C endpoint.
<code>ipv6-loopback</code>	Specifies the name of the IPv6 loopback IP address (associated with this network context) used for this GTP-C endpoint.

Configuring the GTP Proxy Profile

Each GTP Proxy service has a GTP Proxy Profile associated with it that contains common configurable parameters that can be assigned to any number of GTP Proxy services.

Use the following CLI commands to configure the GTP Proxy Profile.

- 1 Configure the GTP Proxy Profile.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway profile
gtp-proxy-profile <gtp-proxy-profile-name>
```

The following table describes the configurable parameters.

Parameter	Description
apn-matching-type	Specifies how the GTP Proxy should use the Network Identifier (NI) and Operator Identifier (OI) parts of the APN Name to authorize session setup requests. This feature simplifies APN configuration by requiring only a single APN to be defined to authorize session setup requests on the GTP Proxy. That single APN is used to authorize session setup requests received with APN names having the same NI but different OI suffixes or no OIs. When the GTP Proxy receives a session setup request, it compares the APN received from the SGSN/SGW with the configured APN (from the APN list) on the GTP Proxy in accordance with the configured apn-matching-type value. If the match is successful, the session setup is successful. ■ exact-match (default) ■ ni-only ■ ni-and-malformed-oi <i>Note: The APN name reported to external devices, such as the OCS and OFCS, is the same as the configured APN name.</i>
clear-subscribers-on-gtpc-path-failure	Specifies the policy for clearing subscribers when the system detects a GTP-C path failure. ■ true – After the system detects the path failure, all calls that go to that remote peer are actively closed. ■ false – All calls are retained until they are manually cleared, cleared by another remote device, or the path recovers and restart processing clears the calls.
reject-gtpc-messages-from-mismatched-src-and-fteid-ip	Specifies the policy for rejecting GTP-C messages received from a source IP address that does not match the sender's F-TEID IP address. This is applicable for non-handover messages that modify or delete a session. ■ true (default) ■ false

Parameter	Description
subscriber-bulk-deactivation-rate-control	Configure subscriber-bulk-deactivation-rate-control parameters, which provide a mechanism to control the rate at which the GTP Proxy deactivates sessions. Operators can control the number of subscriber deactivation requests so as not to overwhelm the peers with multiple requests. When Operators issue a <code>subscriber-clear</code> command, each GTP Proxy service initiates deactivations for all subscribers belonging to that service, based on the configured <code>subscriber-bulk-deactivation-rate-control</code> parameters. Operators can also use the <code>subscriber clear-local</code> command to clear subscribers locally without informing the peers. This command supports the use of filter parameters. For example, the <code>subscriber clear-local gateway-type-sgw</code> command clears all subscribers belonging to all SGWs locally.
is-static	Specifies whether the deactivation rate is a constant value per second or is dynamically learned based on the peer response. ■ true – The GTP Proxy service deactivates subscribers by sending the configured number of deactivation requests every second. ■ false – The GTP Proxy service sends the configured number of initial requests and then dynamically learns the rate by deactivating one subscriber for every subscriber that completes deactivation.
rate	Specifies the maximum number of subscriber deactivation requests to be sent per second when sending a burst of requests. For dynamic rate control, this is the initial burst rate after which the rate is dynamically learned.

2 Associate the GTP Proxy Profile with a GTP Proxy Service interface.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway gtp-proxy
<gtp-proxy-name> gtp-proxy-profile <gtp-proxy-profile-name>
```

Configuring APNs

Use the following CLI command to configure the APNs accessible via the GTP Proxy.

- 1 Configure the APN(s).

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway apn <apn-name>
```

The following table describes the configurable parameters.

Parameter	Description
apn	Specifies the name of the APN.
network-context	Specifies the name of the network context for this APN.
admin-state	Specifies whether to enable or disable the administrative state for this APN. ■ enabled ■ disabled
gtpv0-sessions	Specifies whether to enable or disable support for setting up GTPv0 sessions for an APN. If a GTPv0 session setup request is received when the support is disabled, the GTP Proxy sends a Create PDP Context reject response message with cause set to “resource unavailable”. ■ enabled ■ disabled (default)
allow-dedicated-bearers	Specifies whether to allow the creation of dedicated bearers at the APN level. The MCC also supports a similar setting under the Workflow Control profile, which Operators can use to disable dedicated bearer setup on a per subscriber basis, if it is not set at the APN level. ■ true (default) ■ false

- 2 Associate the APN lists with the GTP Proxy service.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway gtp-proxy
<gtp-proxy-name> apn-list <apn-list-name>
```

Configuring Home and Roaming PLMN IDs

Use the following CLI commands to configure the list of home and roaming PLMN IDs.

- 1 Configure the lists of home and roaming PLMN IDs.

```
an3000-mcm-slot7cpu1(config)# service-construct plmnid-list  
<plmnid-list-name> plmnid <plmnids>
```

- 2 Associate the PLMN ID lists with the GTP Proxy service.

```
an3000-mcm-slot7cpu1(config)# zone <name> gateway gtp-proxy  
<gtp-proxy-name> [home-plmnid-list|roaming-plmnid-list]  
<plmnid-list-name>
```


Configuring a List of PGW/GGSNs

The Workflow Subscriber Analyzer is used to select a PGW/GGSN to which session setup requests are forwarded. When the GTP Proxy receives a session setup request for a UE, it classifies the UE according to configured Subscriber Analyzer keys, such as the IMSI range, APN or UE location. The UE's classification resolves to a list of locally configured PGWs/GGSNs.

The GTP Proxy selects the appropriate PGW/GGSN from the list using a standard or weighted round-robin algorithm and forwards the session setup request. A configured PGW/GGSN will only be used for selection if its link status is up and it supports the GTP version and IP address type that the session is using. To select a GGSN/PGW based on Subscriber Analyzers keys, the `analyzer-type` must be set to `peer-selector`.

When the GTP Proxy selects the PGW/GGSN based on the configured weight assigned to each PGW/GGSN, the weights are added up for all PGWs/GGSNs in the list and a percentage of total weight is used to distribute incoming calls.

Example:

APN Name	PGW IP Address	Weight	Percentage of Total Weight (60)	Distribution of Incoming Calls
consumer.net	1.1.1.1	10	17%	10/60
	1.1.1.2	20	33%	20/60
	1.1.1.3	30	50%	30/60

Use the following CLI commands to configure a list of PGW/GGSNs.

- 1 Configure a list of PGW/GGSNs.

```
an3000-mcm-slot7cpu1(config)# service-construct pdngw-list
<pdngw-list-name> ip-address <ip-address> gtp-version
[gtpv0|gtpv0v1|gtpv0v1v2|gtpv1|gtpv1v2|gtpv2] weight <value>
admin-state [enabled|disabled]
```

where

Parameter	Description
<code>pdngw-list</code>	Specifies the name of the PGW/GGSN list.
<code>ip-address</code>	Specifies the IP address of the PGW/GGSN.
<code>gtp-version</code>	Specifies the GTP version of the interface with the PGW/GGSN.

Parameter	Description
weight	(Optional) Specifies the weight allocated to this PGW/GGSN. The PGW/GGSN with the highest weight receives the most distribution of calls. If weights are specified for the IP addresses configured, the GTP Proxy will use a weighted round-robin algorithm to select the PGW/GGSN. If no weight is configured, the GTP Proxy will use a standard round-robin algorithm.
admin-state	Specifies whether to enable or disable the administrative state for PGW/GGSN. ■ enabled ■ disabled

- Associate the PGW/GGSN list with a Subscriber Analyzer.

```
an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer <subscriber-analyzer-name> analyzer-type peer-selector peer-list <pdngw-list-name>
```

- Configure the keys used for PGW/GGSN selection.

```
an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer <subscriber-analyzer-name> key <key-name> <your-list-of-keys> <key-values>
```

where

Parameter	Description
subscriber-analyzer	Specifies the Subscriber Analyzer name.
key	Specifies the list of analyzed keys. A key can have multiple attributes. You can configure multiple keys, each identified by name. If any key matches, the Subscriber Analyzer is selected. A key can specify multiple triggers, all of which must match for a key to match. <i>Note: For a list of the supported Subscriber Analyzer keys used for PGW/GGSN and Workflow selection, see Subscriber Analyzer Keys (page 1-8).</i>

Configuring Workflow Components and Services

Perform the following steps to configure Workflow components and services on the GTP Proxy. Workflow defines the treatment of a service flow across various services. For more information, see [Workflow Profiles \(page 1-7\)](#).

For more information on configuring Workflow components and services, see the *Affirmed Networks Gateway Administration Guide* and *Affirmed Networks Value Added Services Administration Guide*.

- 1 Create an online charging service instance (Gy interface) on the GTP Proxy. For more information, see [Online Charging \(page 1-10\)](#).
- 2 Create a Workflow Data profile to specify the services to provide for the PDN session and controls resource utilization by the session. When the Workflow is established, each associated subscriber's traffic flow is steered to the collection of associated services. Currently, only the charging service can be added in the Data profile.

```
an3000-mcm-slot7cpu1(config)# workflow data-profile
<data-profile-name> charging <charging-instance-name>
```

- 3 Create a Workflow Control profile to specify the back-end interfaces for charging and accounting.

When APN mapping is used, the Workflow Control profile is associated with an APN name to which the received APN name is mapped. Using the `apn-reporting` parameter under the Workflow Control profile, Operators can specify the APN name that will be forwarded to the PGW/GGSN and to external servers, such as the OCS (Gy interface) and OFCS (Ga interface). For more information APN mapping, see [APN Mapping \(page 1-18\)](#).

```
an3000-mcm-slot7cpu1(config)# workflow control-profile
<control-profile-name> default-offline-interface
<offline-interface-name> default-online-interface
<online-interface-name> apn-mapping-control explicit mapped-apn-name
<apn-name> apn-reporting original
```

where

Parameter	Description
<code>control-profile</code>	Specifies the Workflow Control profile name.
<code>default-offline-interface</code>	Specifies the offline charging interface name (Ga interface).
<code>default-online-interface</code>	Specifies the online charging interface name (Gy interface).
<code>apn-mapping-control</code>	Specifies the control for mapping to an APN. In this case, select <code>explicit</code> to map to the APN specified by the <code>mapped-apn-name</code> field.

Parameter	Description
mapped-apn-name	Specifies the mapped APN name to which the received APN name is mapped. The APN must already exist. See Configuring APNs (page 2-15) .
apn-reporting	Specifies the APN to use when reporting the APN to external devices, such as the OCS and OFCS. In this case, select <code>original</code> .

4 Associate the Control and Data profiles with a Subscriber Analyzer.

```
an3000-mcm-slot7cpu1(config)# workflow subscriber-analyzer
<subscriber-analyzer-name> analyzer-type workflow-profile priority
<priority-number> key <key-name> <your-list-of-keys> <key-values>;
action data profile <data-profile-name> control-profile
<control-profile-name>
```

where

Parameter	Description
subscriber-analyzer	Specifies the Subscriber Analyzer name.
analyzer-type	Specifies the type of Subscriber Analyzer. In this case, select <code>workflow-profile</code> .
event	Specify whether to associate the Workflow with Bearer Setup.
geo-redundant-group	Specifies the Geographic Redundancy group name.
priority	Specifies the priority. The priority must be unique.
key	Specifies the list of analyzed keys. A key can have multiple attributes. You can configure multiple keys, each identified by name. If any key matches, the Subscriber Analyzer is selected. A key can specify multiple triggers, all of which must match for a key to match. <i>Note:</i> For a list of the supported Subscriber Analyzer keys used for PGW/GGSN and Workflow selection, see Subscriber Analyzer Keys (page 1-8) .
apn-name	Specifies the Access Point Name.
data profile	Specifies the Workflow Data profile. <i>Note:</i> Currently, only the charging service can be added in the Data profile.
control-profile	Specifies the Workflow Control profile identifying the external identity providing the Workflow Data profile.

Statistics and Alarms

This section describes how to display GTP Proxy statistical information, monitor the operational status of the GTP peer and describes the alarm notification that occurs when the remote GTP peer goes down.

- [GTP Proxy Call Performance Statistics \(page 3-2\)](#)
- [Interface Statistics \(page 3-3\)](#)
- [Procedure-Level Statistics \(page 3-3\)](#)
- [GTP Peer Status \(page 3-3\)](#)
- [Alarms \(page 3-4\)](#)

For more information on viewing statistics, see *Acuitas Performance Statistics*.

For information on viewing Key Performance Indicators (KPIs), see the *Acuitas User's Guide*.

GTP Proxy Call Performance Statistics

To view GTP Proxy call performance statistics with the MCC CLI, use the following command.

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics  
[cluster-level|grid-level|node-level] call-performance gtpproxy  
<gtp-proxy-name>
```

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics  
[cluster-level|grid-level|node-level] call-performance gtpproxy-apn  
<apn-name>
```

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics  
[cluster-level|grid-level|node-level] call-performance  
gtpproxy-apn-qci-arp-pl
```

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics  
[cluster-level|grid-level|node-level] call-performance  
gtpproxy-qci-arp-pl
```

To view these statistics with Acuitas Performance Management, see *Acuitas Performance Statistics*.

Interface Statistics

You can display statistical information for interface communication messages for the grid, cluster and node level for a specific GTP peer as shown in the following examples.

Grid Level

Use the following CLI command to display the interface GTP grid-level statistics for a specific peer.

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics grid-level
gtp peer <peer-name>
```

Cluster Level

Use the following CLI command to display the interface GTP cluster-level statistics for a specific peer.

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics
cluster-level gtp peer <peer-name> <cluster-id>
```

Node Level

Use the following CLI command to display the interface GTP node-level statistics for a specific peer.

```
an3000-mcm-slot7cpu1# show zone <name> gateway statistics node-level
gtp peer <peer-name> <cluster-id> <slot>
```

Procedure-Level Statistics

Use the following CLI commands to display procedure-level statistics:

```
an3000-mcm-slot7cpu1# show zone default gateway statistics
[cluster-level|grid-level|node-level] procedure gtpproxy
<gtp-proxy-name>
```

```
an3000-mcm-slot7cpu1# show zone default gateway statistics
procedure-duration gtpproxy <gtp-proxy-name>
```

GTP Peer Status

Use the following CLI command to display the status information of a GTP peer.

```
an3000-mcm-slot7cpu1# show zone <name> gateway gtp-peer-status
```

Use the following CLI command to display the status information of a GTP peer within a network context.

```
an3000-mcm-slot7cpu1# show zone <name> gateway gtp-peer-status
<gateway-name> <peer-ip> <network-context>
```

Alarms

A major-level trap similar to the following occurs when a GTP peer goes down. The alarm clears after connectivity is restored. For more information on alarms, see the *Affirmed Networks SNMP MIB and Alarm Reference Guide*.

anGtpPeerDown

Description: GTP peer down. This indicates that the remote GTP peer is not responding to any GTP messages sent to it. If the peer is statically provisioned, the severity is Major. If the peer is dynamic, the severity is Info.

Trap ID: 60

Status: Current

Severity: Info, Major

Auto Clearing: No

Potential Impact: Sessions that go to the remote peer may be cleared, resulting in loss of data connectivity for the affect UEs

Corrective Action: Verify that the remote peer is up and running. Verify that the remote peer is reachable. Verify that the local GTP endpoint is correctly provisioned and running normally.

Note: When the interface connection to a dynamic GTP peer goes down, the system generates an Info alarm.
